

Lithium acetate/PEG transformation of snowflake yeast

Adapted from the lab compendium Snowflake Yeast Protocols — Ratcliff Lab / MuLTEE (V2).

Reagents (per sample)

Reagent	Volume
PEG (50% w/w)	260 μ L
LiOAc (1 M)	36 μ L
ssDNA (2 mg/mL, freshly boiled)	50 μ L
Digested plasmid/DNA	20 μ L
ddH ₂ O	34 μ L

Pre-culture

1. Inoculate 2 mL from an overnight culture into 10 mL fresh growth medium.
2. Grow the pre-culture for 3 hours.

Alternative: inoculate 1 mL from an overnight culture and grow for 4 hours instead of steps 1-2.

Cell prep

1. Transfer 1 mL of pre-culture into a labeled 1.5 mL tube (one tube per line/population). Vortex briefly before use.
2. Centrifuge at 8,000 rpm for 1 min.
3. Confirm pellet volume reaches approximately the 100 μ L mark (minimum halfway to 100 μ L). If insufficient, add another 1 mL of culture to the same tube and re-spin.
4. Remove supernatant without disturbing the pellet.
5. Add 100 μ L water and pipette up and down to break up clusters until the mixture is evenly suspended (light cream color, no visible individual colonies). Later-generation clusters require more vigorous pipetting.
6. Add 900 μ L water (1 mL total), centrifuge (8,000 rpm, 1 min), remove supernatant.
7. Wash with 1 mL water a second time (centrifuge, discard supernatant).
8. Wash with 1 mL 0.1 M LiOAc twice (dilute stock, not the 1 M stock used in [[> Assembling the transformation mix]]). Each wash: add liquid, centrifuge, discard supernatant.
9. Resuspend the final pellet in 100 μ L 0.1 M LiOAc, pipetting to break up clusters.
10. Centrifuge and remove supernatant thoroughly.

ssDNA prep (concurrent with LiOAc washes)

1. Heat salmon sperm DNA (ssDNA) to 95 °C for 5 min.

2. Set heat block to 42 °C immediately after.
3. Mark the ssDNA tube each time it is boiled; do not reboil the same aliquot more than 3-4 times.
4. Recover boiled ssDNA on ice quickly before using at room temperature
5. Use ssDNA within ~30 min of boiling.

Assembling the transformation mix

Add to each cell pellet, in order:

1. **260 µL PEG (50% w/w)** — pipette slowly due to viscosity.
2. **36 µL 1 M LiOAc** — mix/swirl until uniformly cloudy.
 - Note specialized mixing technique:
 - Pipette up cells from bottom of solution and release at the top.
 - Swirl pipette around the sides of the tube on the way back down to the bottom to mix.
 - Repeat until large clusters crushed and solution uniform.
3. **50 µL freshly boiled ssDNA** (vortex briefly before adding).
4. **At least 20 µL DNA solution** (plasmid, PCR product, etc.). For CRISPR: 10 µL Cas9 construct + 10 µL insert DNA. Mix before adding.
5. **ddH₂O** as necessary so ddH₂O + DNA solution add to **54 µL**

Heat shock and recovery

1. Incubate at **30 °C for 30 min** with rotation.
2. Heat-shock at **42 °C for 17 min** (heat block, ~350 rpm if available).
3. Centrifuge at 8,000 rpm for 2 min; remove all supernatant.
4. Resuspend in **1.0 mL YPD**, transfer to a fresh 2 mL tube, parafilm the lid closed to prevent lid popping open.
5. Recover at **30 °C for 4 hours**.
6. Plate the desired volume (typically 100-500 µL) on selective medium.